Assignment_03

1

读取文件信息

xarray.Dataset

▸ Dimensions:	(**time**: 210, bnds: 2, **lat**: 36, **lon**: 72, pressure: 10)		
▾ Coordinates:			
time	(time)	datetime64[ns]	2003-01-16T12:00:00 ... 2020-0...
lat	(lat)	float64	-87.5 -82.5 -77.5 ... 82.5 87.5
lon	(lon)	float64	-177.5 -172.5 ... 172.5 177.5
▾ Data variables:			
time_bnds	(time, bnds)	datetime64[ns]	...
lat_bnds	(lat, bnds)	float64	...
lon_bnds	(lon, bnds)	float64	...
pre	(pressure)	float64	...
pre_bnds	(pressure, bnds)	float64	...
land_fraction	(lat, lon)	float64	...
xch4	(time, lat, lon)	float32	...
xch4_nobs	(time, lat, lon)	float64	...
xch4_stderr	(time, lat, lon)	float32	...
xch4_stddev	(time, lat, lon)	float32	...
column_averag...	(time, pressure, lat, lon)	float32	...
vmr_profile_ch...	(time, pressure, lat, lon)	float32	...
▸ Attributes: (28)			

提取甲烷数据

```
<xarray.DataArray 'xch4' (time: 210, lat: 36, lon: 72)> Size: 2MB
[544320 values with dtype=float32]
Coordinates:
  * time     (time) datetime64[ns] 2kB 2003-01-16T12:00:00 ... 2020-06-16
  * lat      (lat) float64 288B -87.5 -82.5 -77.5 -72.5 ... 72.5 77.5 82.5 87.5
  * lon      (lon) float64 576B -177.5 -172.5 -167.5 ... 167.5 172.5 177.5
Attributes:
    standard_name:  dry_atmosphere_mole_fraction_of_methane
    long_name:      column-average dry-air mole fraction of atmospheric methane
    units:          1
    cell_methods:   time: mean
    fill_value:     1e+20
    comment:        Satellite retrieved column-average dry-air mole fraction ...
```

将无效数据和缺测数据转化成 NAN

单位转化成 ppb

```
<xarray.DataArray 'xch4' (time: 210, lat: 36, lon: 72)> Size: 2MB
array([[[nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        ...,
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan]],

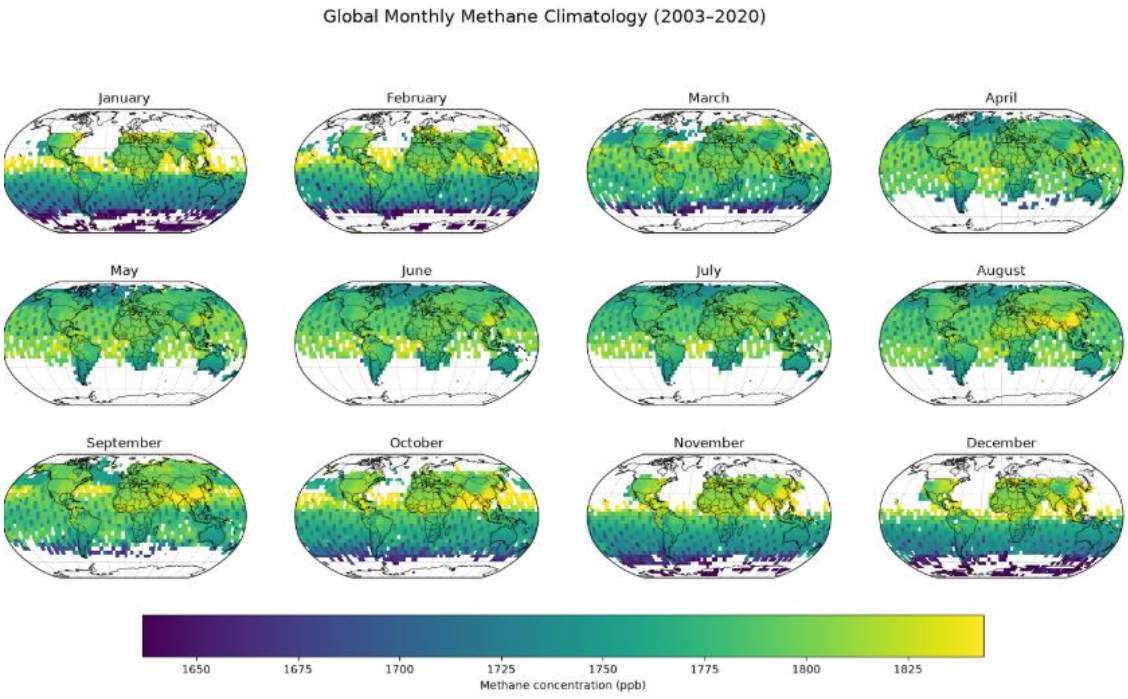
       [[nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        ...,
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan]],

       [[nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        ...,
...
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan]],

       [[nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        ...,
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan]],

       [[nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        ...,
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan],
        [nan, nan, nan, ..., nan, nan, nan]]],
      shape=(210, 36, 72), dtype=float32)
Coordinates:
  * time     (time) datetime64[ns] 2kB 2003-01-16T12:00:00 ... 2020-06-16
  * lat      (lat) float64 288B -87.5 -82.5 -77.5 -72.5 ... 72.5 77.5 82.5 87.5
  * lon      (lon) float64 576B -177.5 -172.5 -167.5 ... 167.5 172.5 177.5
Attributes:
    standard_name:  dry_atmosphere_mole_fraction_of_methane
    long_name:      column-average dry-air mole fraction of atmospheric methane
    units:          ppb
    cell_methods:   time: mean
    fill_value:     1e+20
    comment:        Satellite retrieved column-average dry-air mole fraction ...
```

1.1 气候态结果图

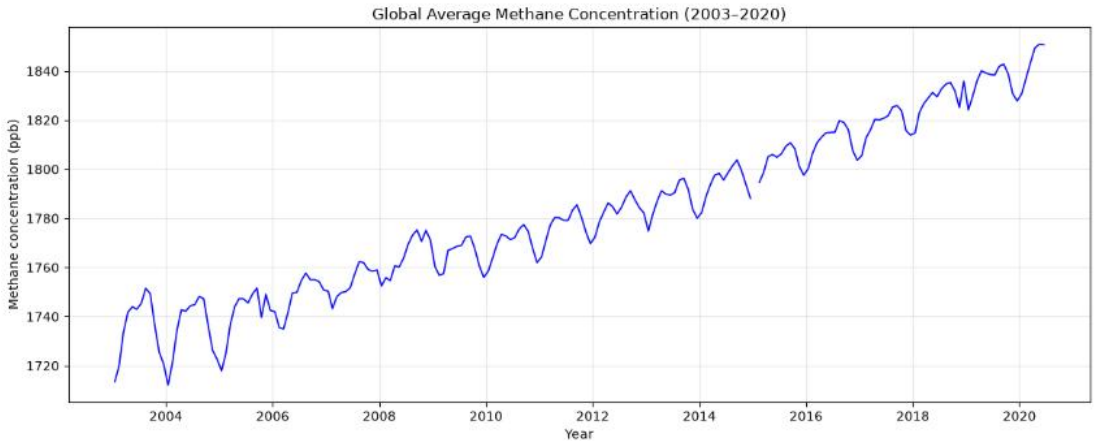


1.2

计算权重

```
<xarray.DataArray 'xch4' (time: 210)> Size: 2kB
array([1713.49139949, 1719.92258196, 1733.51969734, 1741.75989025,
       1743.96019217, 1742.9729384 , 1745.33766449, 1751.44989244,
       1749.33122929, 1736.69786687, 1725.54902848, 1720.73877732,
       1712.14989303, 1721.31922389, 1734.21696401, 1742.69156058,
       1742.19190561, 1744.34731695, 1744.86202533, 1748.20243246,
       1747.19370063, 1736.42159882, 1726.23678054, 1722.66895711,
       1717.89884968, 1724.99139444, 1736.46319431, 1743.95731199,
       1747.27492667, 1747.21798719, 1745.53864271, 1749.20921664,
       1751.61246948, 1739.70947221, 1749.01717181, 1742.50406944,
       1741.97405441, 1735.54960511, 1734.91747187, 1741.63596122,
       1749.59818772, 1749.87683157, 1754.65085402, 1757.66751662,
       1755.00692358, 1755.05138001, 1754.14885375, 1750.79010476,
       1750.31263013, 1743.26036816, 1748.08683102, 1749.77185635,
       1750.14281217, 1751.67632611, 1757.21801887, 1762.33249918,
       1761.94947028, 1759.18897415, 1758.41061011, 1758.96427642,
       1752.41883398, 1755.83524004, 1754.60214721, 1760.61365359,
       1760.19824189, 1763.79456024, 1769.37185731, 1773.17500662,
       1775.25740157, 1770.6275097 , 1775.08721203, 1771.10447057,
       1760.30775106, 1756.7811168 , 1757.43591145, 1766.93543362,
       1767.67800761, 1768.57234789, 1768.93528938, 1772.3721849 ,
...
       1782.31425798, 1788.82118903, 1793.57979778, 1797.60554929,
       1798.29985409, 1795.59703437, 1798.60874718, 1801.48742962,
       1803.7554556 , 1799.39514508, 1793.74936103, 1788.10109981,
                nan, 1794.63004394, 1798.60970325, 1805.05997529,
       1805.9906459 , 1804.84593266, 1806.28345515, 1809.36781278,
       1810.77207028, 1808.30077255, 1801.08374363, 1797.56033736,
       1800.27906221, 1806.58414954, 1810.79523273, 1813.01156741,
       1814.79011287, 1814.95693276, 1815.05174625, 1819.78348798,
       1818.98307738, 1816.09640897, 1807.56827831, 1803.65470411,
       1805.47657848, 1812.77146332, 1815.91861821, 1820.2737877 ,
       1820.0752333 , 1820.72220054, 1821.78606619, 1825.32020498,
       1825.87373471, 1823.75954852, 1815.68626534, 1813.87041815,
       1814.7417724 , 1823.00065679, 1826.6100676 , 1828.94949744,
       1831.25563653, 1829.46398825, 1832.7137426 , 1834.65234811,
       1835.29877241, 1831.83591507, 1825.08162854, 1835.77964771,
       1824.10103001, 1829.70873038, 1835.87827931, 1840.04363438,
       1839.13364002, 1838.5206176 , 1838.34857391, 1841.93227343,
       1842.78298496, 1838.71381641, 1830.73109037, 1827.79833777,
       1830.68806564, 1837.02947938, 1843.36253839, 1849.38268164,
       1850.89249013, 1850.67171403])
Coordinates:
  * time     (time) datetime64[ns] 2kB 2003-01-16T12:00:00 ... 2020-06-16
Attributes:
    standard_name:  dry_atmosphere_mole_fraction_of_methane
    long_name:      column-average dry-air mole fraction of atmospheric methane
    units:          ppb
    cell_methods:   time: mean
    fill_value:     1e+20
    comment:        Satellite retrieved column-average dry-air mole fraction ...
```

全球平均甲烷浓度的时间序列图

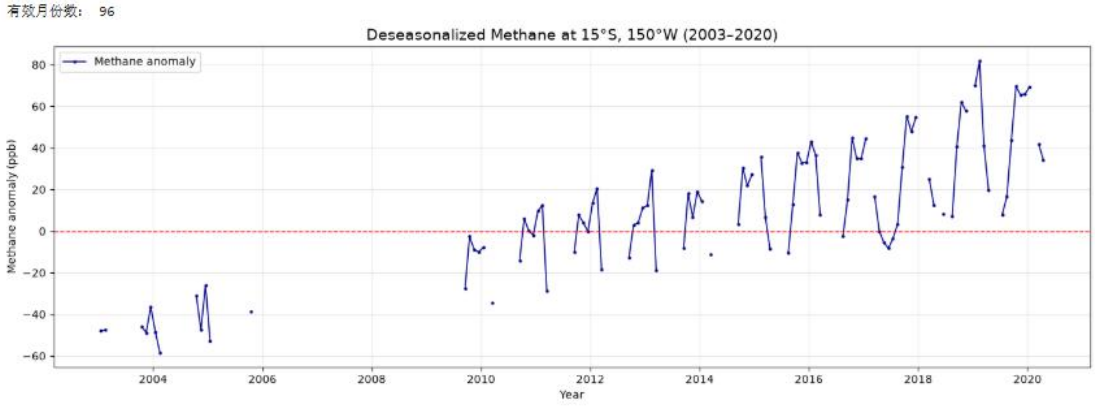


全球平均甲烷总体明显上升，同时叠加年内季节波动。约 2003-2006 年增长相对较慢，之后重新加速，2014 年以后上升更明显。长期上升及季节起伏趋势与 NASA 趋势一致

1.3

使用 nearby 方法提取目标位置附件的网格数据

至少有 2 个有效网格数据才保留，否则输出为 nan



2

读取文件信息

xarray.Dataset

- Dimensions: (**time**: 684, **lat**: 89, **lon**: 180)
- Coordinates:

time	(time)	datetime64[ns]	1960-01-15 ... 2016-12-15
lat	(lat)	float32	-88.0 -86.0 -84.0 ... 86.0 88.0
lon	(lon)	float32	0.0 2.0 4.0 ... 354.0 356.0 358.0

- Data variables:

sst	(time, lat, lon)	float32	...

- Attributes:

Conventions :	IRIDL
source :	https://iridl.ldeo.columbia.edu/SOURCES/.NOAA/.NCDC/.ERSST/.version3b/.sst/
history :	extracted and cleaned by Ryan Abernathey for Research Computing in Earth Science

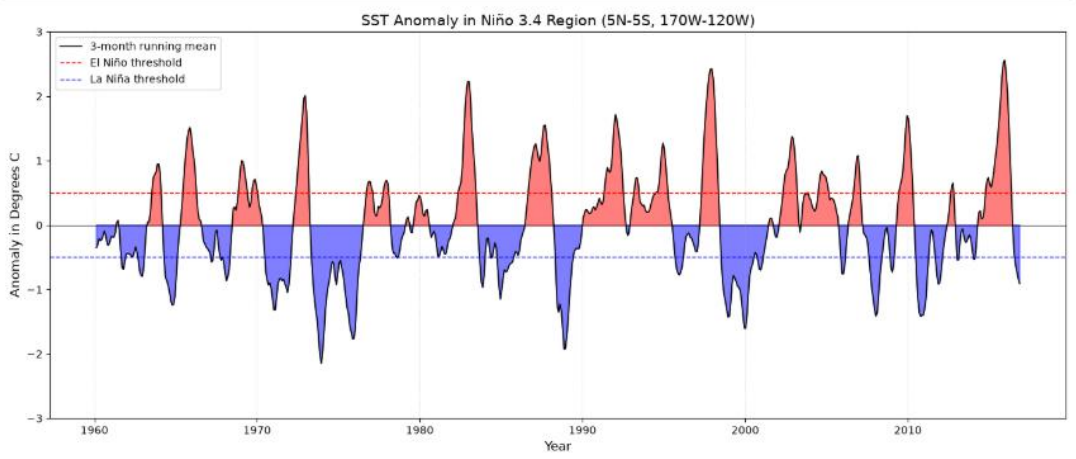
2.1 月气候态数据

month	SST (°C)
1	26.567789
2	26.742273
3	27.239603
4	27.693741
5	27.795214
6	27.597757
7	27.198917
8	26.824203
9	26.737827
10	26.717142
11	26.693287
12	26.613092

异常值数据

```
<xarray.DataArray 'sst' (time: 684)> Size: 3kB
array([-3.19671631e-01, -4.68568802e-01, -2.68173218e-01, -1.87025070e-01,
       -1.77633286e-01, -3.57751846e-01, -1.42024994e-01,  1.46160126e-02,
       -1.52326584e-01, -3.79961014e-01, -3.60948563e-01, -2.08705902e-01,
       -1.91469193e-01, -1.34254456e-01, -2.40343094e-01, -1.85152054e-01,
        5.19752502e-02,  2.39465714e-01, -7.02838898e-02, -4.18712616e-01,
       -7.78390884e-01, -7.92451859e-01, -4.96280670e-01, -4.17322159e-01,
       -4.44835663e-01, -4.56037521e-01, -4.36143875e-01, -4.66995239e-01,
       -5.64611435e-01, -4.26933289e-01, -2.76754379e-01, -3.17255020e-01,
       -6.73257828e-01, -6.66168213e-01, -8.17825317e-01, -8.65585327e-01,
       -7.01501846e-01, -3.57101440e-01,  6.96544647e-02,  1.73439026e-01,
       -9.07344818e-02,  7.68814087e-02,  6.87768936e-01,  7.95410156e-01,
        7.13079453e-01,  8.34375381e-01,  9.08702850e-01,  1.08613205e+00,
        8.51739883e-01,  5.23542404e-01, -1.30517960e-01, -6.34527206e-01,
       -8.62230301e-01, -9.46153641e-01, -7.84320831e-01, -1.09451294e+00,
       -1.25469017e+00, -1.18999290e+00, -1.29210663e+00, -1.20490646e+00,
       -7.83042908e-01, -3.68721008e-01, -1.12571716e-01, -7.66201019e-02,
        2.68445969e-01,  5.91285706e-01,  8.12198639e-01,  1.23831367e+00,
        1.27783585e+00,  1.52458572e+00,  1.58341789e+00,  1.44135284e+00,
        1.17611122e+00,  9.12860870e-01,  9.74102020e-01,  5.48101425e-01,
       -1.27288818e-01,  1.68792725e-01,  2.08026886e-01, -1.75617218e-01,
...
       -1.43023300e+00, -1.45682907e+00, -1.37612152e+00, -1.35096741e+00,
       -1.47339058e+00, -1.05754662e+00, -8.56292725e-01, -4.86480713e-01,
       -1.43394470e-01,  4.31327820e-02, -1.12285614e-01, -3.01332474e-01,
       -5.81214905e-01, -7.46570587e-01, -9.97346878e-01, -1.00209427e+00,
       -7.01263428e-01, -5.66427231e-01, -3.01498413e-01, -1.52091980e-01,
       -8.46176147e-02,  7.59658813e-02,  1.72382355e-01,  3.80504608e-01,
        8.36627960e-01,  6.27626419e-01,  4.97903824e-01, -3.83304596e-01,
       -6.95823669e-01, -5.49386978e-01, -3.67597580e-01,  9.52854156e-02,
       -3.93810272e-02, -2.35090256e-01, -2.66677856e-01, -2.97039032e-01,
       -2.14778900e-01, -8.15753937e-02, -1.58256531e-01, -3.91790390e-01,
       -5.38154602e-01, -6.71583176e-01, -3.81248474e-01,  2.23903656e-02,
        3.64423752e-01,  2.08328247e-01,  8.79955292e-02, -1.89590454e-03,
        2.54854202e-01,  5.18747330e-01,  8.60292435e-01,  7.23615646e-01,
        6.23155594e-01,  5.44227600e-01,  5.93421936e-01,  9.04737473e-01,
        1.00395966e+00,  1.14287186e+00,  1.40430069e+00,  1.65575027e+00,
        1.91502953e+00,  2.32413101e+00,  2.70273590e+00,  2.52209282e+00,
        2.46219063e+00,  2.14323616e+00,  1.61821175e+00,  1.15107727e+00,
        4.64040756e-01, -1.12422943e-01, -5.67667007e-01, -6.47998810e-01,
       -6.40214920e-01, -8.96921158e-01, -9.71063614e-01, -8.55525970e-01],
      dtype=float32)
Coordinates:
  * time     (time) datetime64[ns] 5kB 1960-01-15 1960-02-15 ... 2016-12-15
    month    (time) int64 5kB 1 2 3 4 5 6 7 8 9 10 11 ... 3 4 5 6 7 8 9 10 11 12
Attributes:
    pointwidth:          1.0
    valid_min:           -3.0
    valid_max:           45.0
    units:               degree_Celsius
    long_name:           Extended reconstructed sea surface temperature
    standard_name:       sea_surface_temperature
    iridl:hasSemantics:  iridl:SeaSurfaceTemperature
```

2.2 绘图



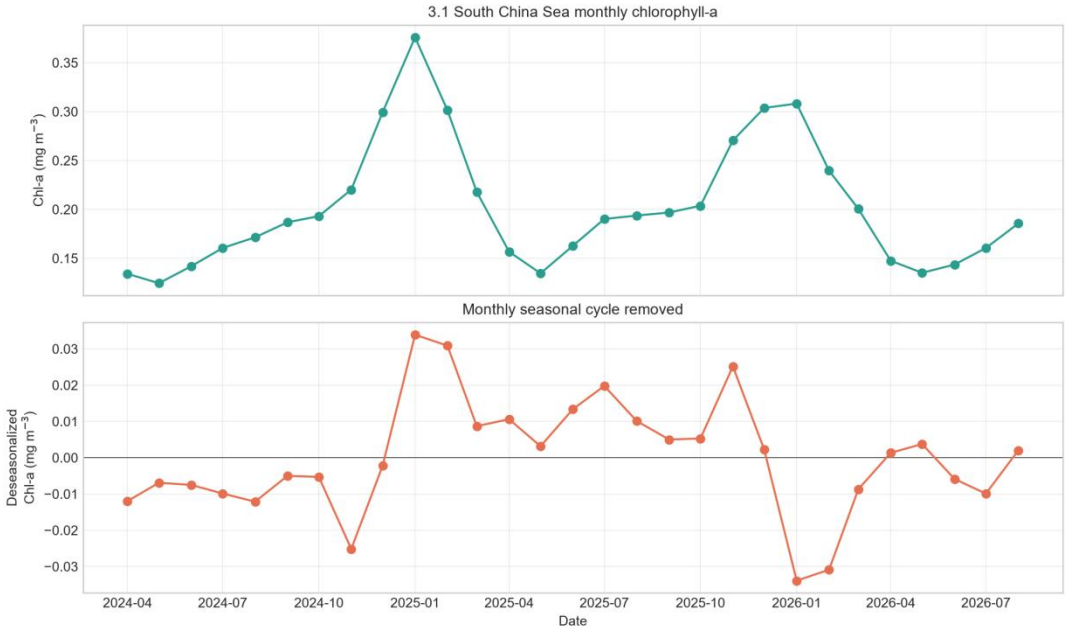
3

使用 PACE OCI L3 Chl-a 0.1° 月度数据，数据可在 NASA Ocean Color 下载

3.1 去除月季节循环的时间序列

选择南海区域 5°N-25°N、105°E-121°E。由于叶绿素浓度分布明显右偏，先对有效值取自然对数，再使用纬度余弦作为面积权重计算区域几何平均。

将每个月的叶绿素浓度减去相同日历月的多年平均值，得到去季节序列。



3.2 使用 PACE 叶绿素数据绘制五种不同图形

图 1 单月全球叶绿素空间分布

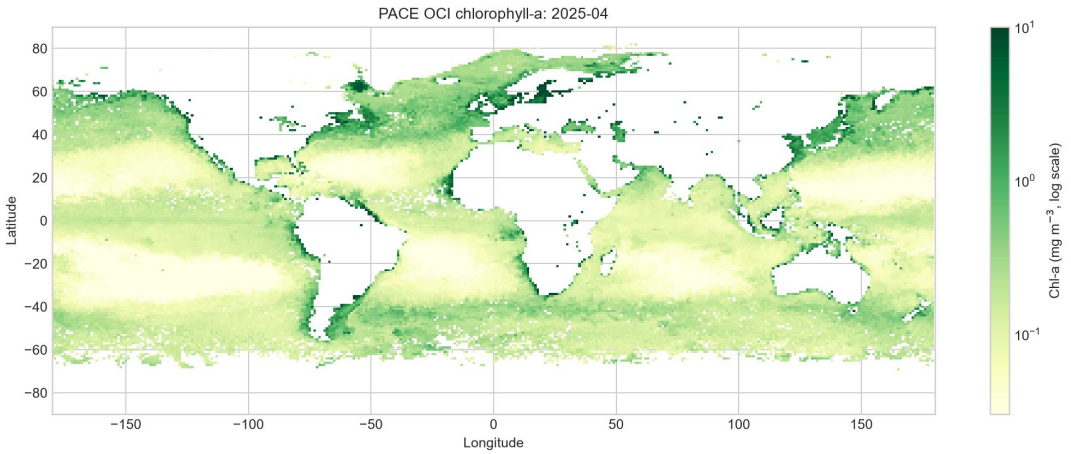


图 2　南海月气候态

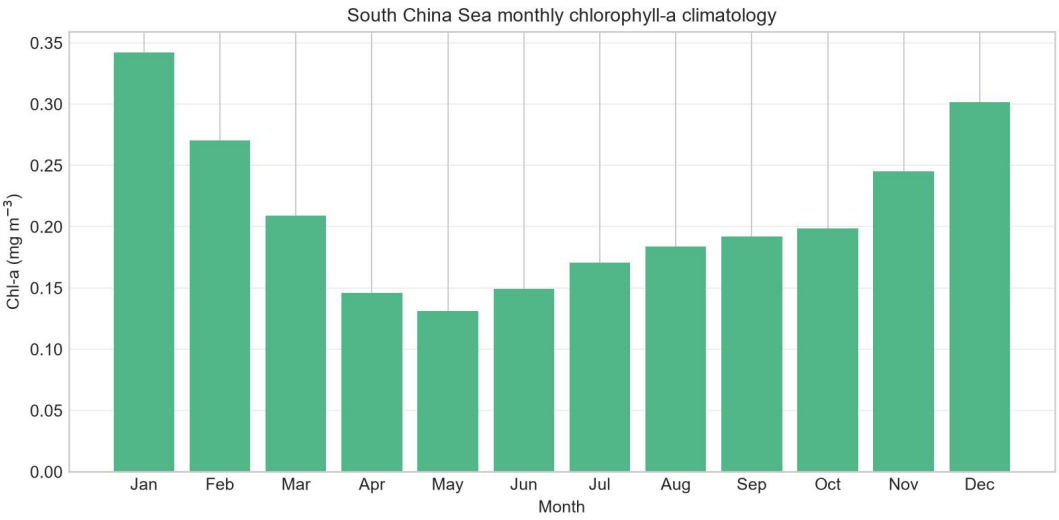


图 3　四季平均空间分布

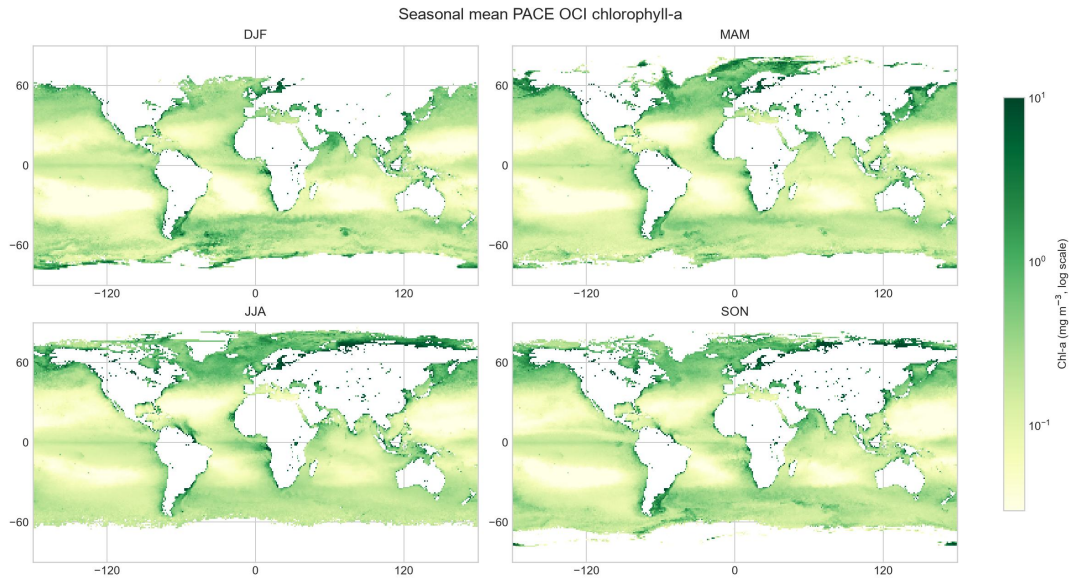


图 4　纬度-时间图

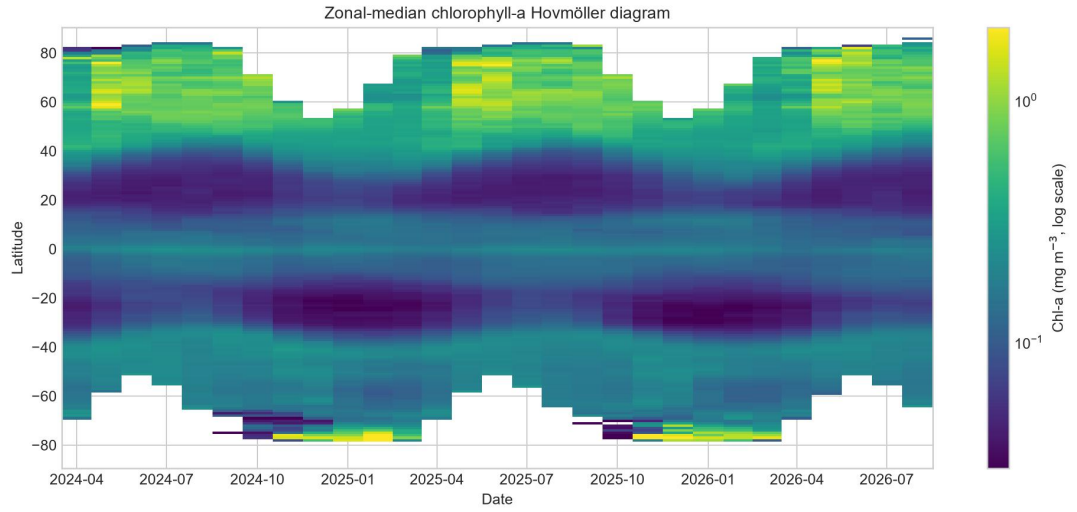
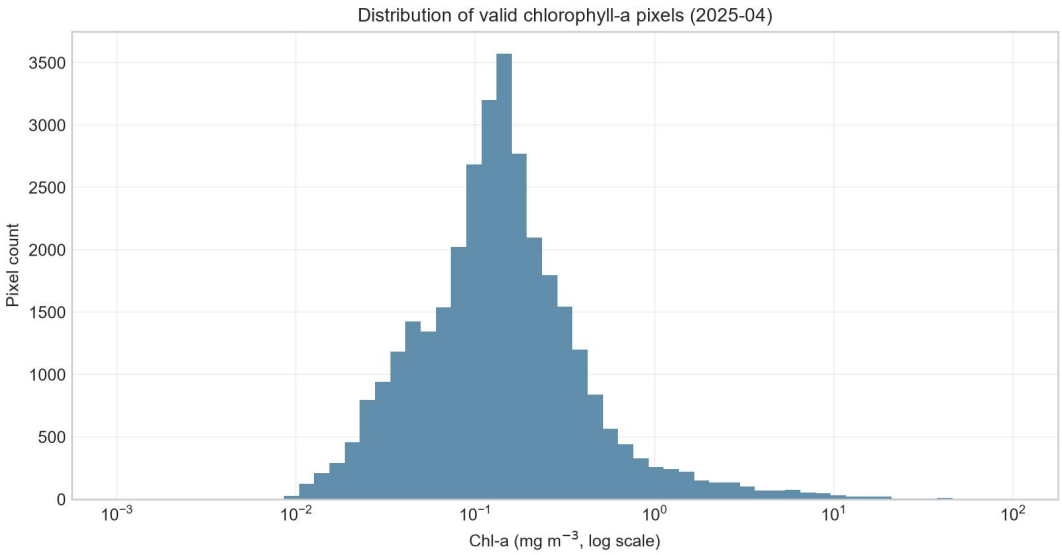


图 16　各纬度叶绿素经度中位数随时间的变化

图 5　叶绿素浓度频率分布


Distribution of valid chlorophyll-a pixels (2025-04)
Pixel count
3500
3000
2500
2000
1500
1000
500
0
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
Chl-a (mg m^{-3}, log scale)